

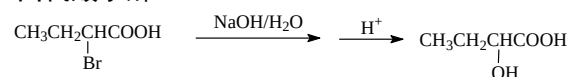


第十三章 取代酸 和 β -二羰基化合物

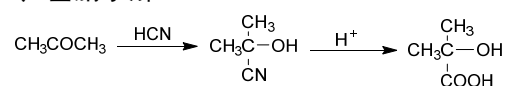
Substituted Carboxylic Acids and β -Dicarbonyl Compounds

羟基酸的制备

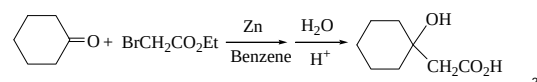
- 卤代酸水解



- 羟基腈水解



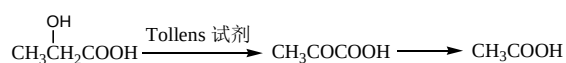
- Reformatsky反应



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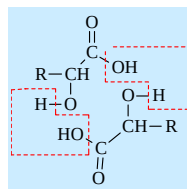
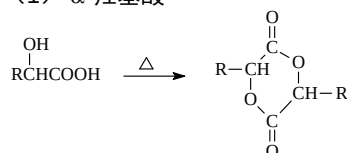
羟基酸的化学反应

- 氧化反应

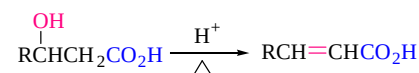


- 脱水反应

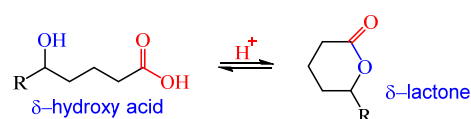
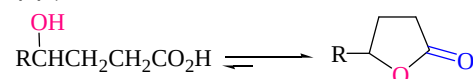
(1) α -羟基酸



(2) β -羟基酸

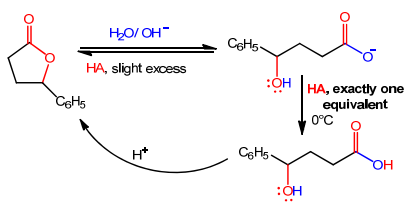
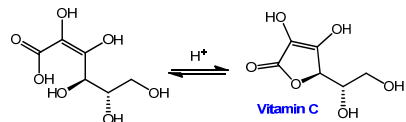


(3) γ -羟基酸

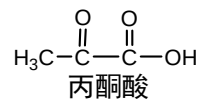
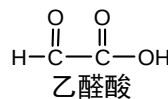


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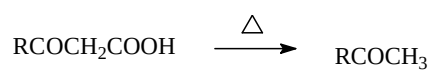
Lactones



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 α -二羰基化合物

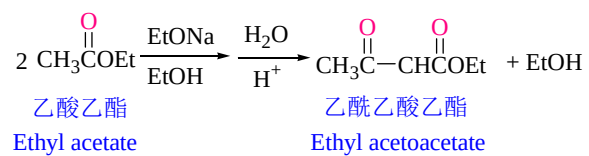
- 酸性大于羧酸
- α 或 β -酮酸受热易脱羧



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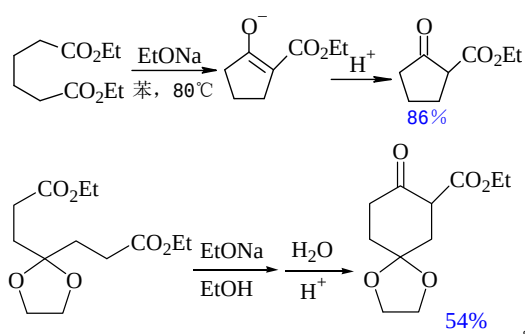
 β -二羰基化合物1. β -酮酸酯的形成 (机理见12章)

Claisen 酯缩合



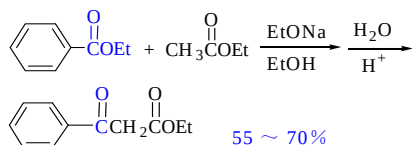
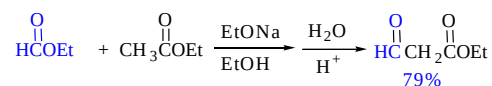
7

Dieckmann反应



8

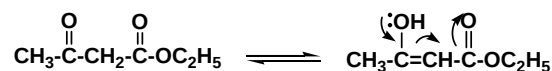
不对称二酯和混合酯的缩合



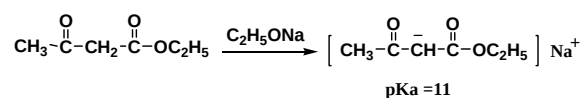
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β -二羰基化合物

2. β -酮酸酯的互变异构现象



亚甲基活泼氢的性质：酸性



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Keto and Enol Tautomers

酮式和烯醇式虽然共存于一个平衡体系中，但在绝大多数情况下，酮式是主要的存在形式。但是随着 α 氢的活泼性增大，失去氢后形成的碳负离子的稳定性增大，烯醇式也可能成为平衡体系中的主要存在形式：

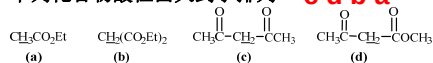
化合物	酮式和烯醇式的互变异构	烯醇式含量 (%)
丙酮	$\text{CH}_3\text{COCH}_3 \rightleftharpoons \text{CH}_2=\text{C}(\text{OH})\text{CH}_3$	0.00015
丙二酸二乙酯	$\text{C}_2\text{H}_5\text{OOCCH}_2\text{OC}_2\text{H}_5 \rightleftharpoons \text{C}_2\text{H}_5\text{OOCCH}(\text{OH})\text{COC}_2\text{H}_5$	0.0077
乙酰乙酸乙酯	$\text{CH}_3\text{COCH}_2\text{COC}_2\text{H}_5 \rightleftharpoons \text{CH}_3\text{C(OH)=CHCOC}_2\text{H}_5$	7.3
2, 4-戊二酮	$\text{CH}_3\text{COCH}_2\text{CCH}_3 \rightleftharpoons \text{CH}_3\text{C(OH)=CHCCH}_3$	76.5
1, 1, 1-三氟-2, 4-戊二酮	$\text{CH}_3\text{COCF}_3\text{CH}_2\text{CCH}_3 \rightleftharpoons \text{CH}_3\text{C(OH)=CHCF}_3$	99

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测试

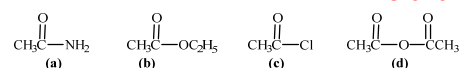
1) 下列化合物酸性由大到小排列

c d b a

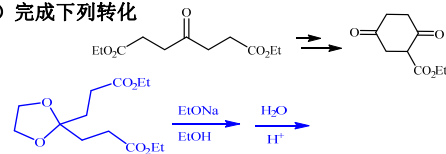


2) 将下列化合物胺解，活性由大到小排列

c d b a

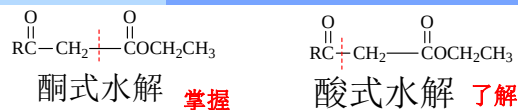


3) 完成下列转化

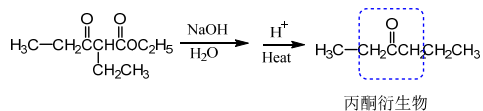


12

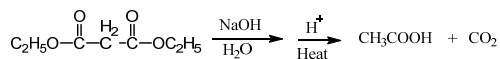
β-酮酸酯的水解



酮式分解: β-羰基乙酸酯在稀碱中水解失羧转化为酮的反应:



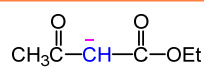
丙二酸二乙酯经酮式分解得乙酸或乙酸的衍生物:



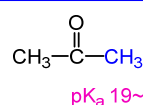
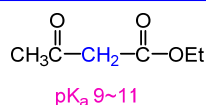
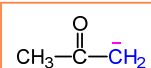
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三乙以及酮式分解的应用

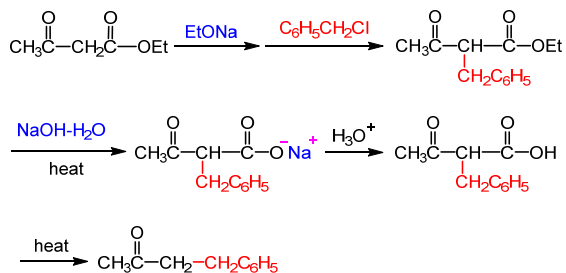
合成烷基酮:



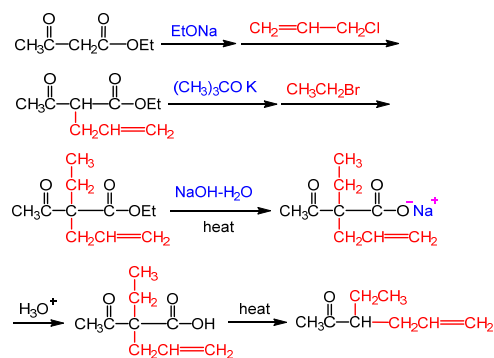
is the synthetic equivalent of



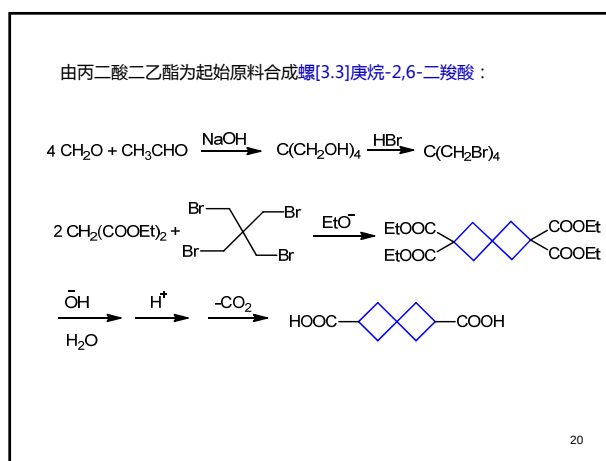
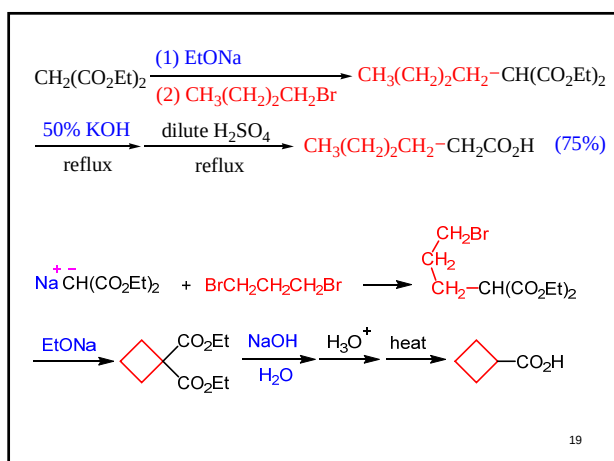
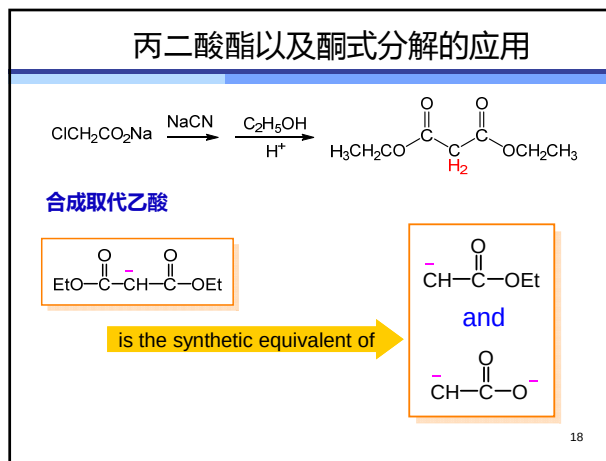
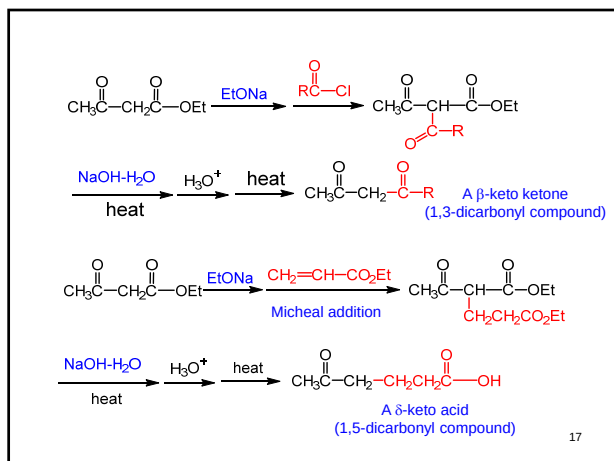
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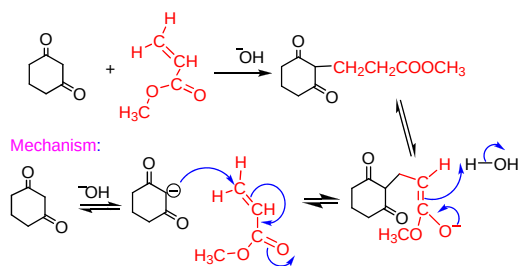
15



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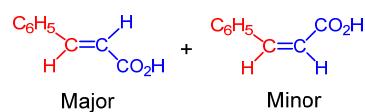
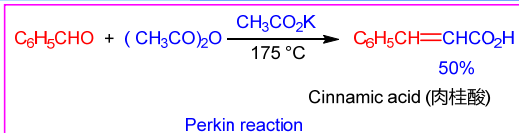


Micheal 加成反应



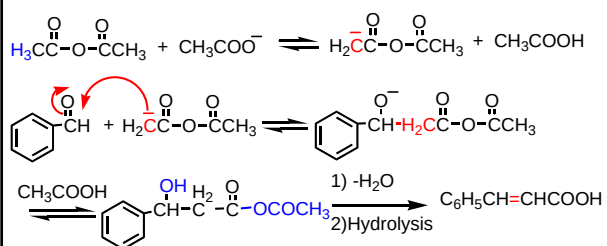
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Perkin 反应



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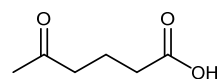
反应机理



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测试

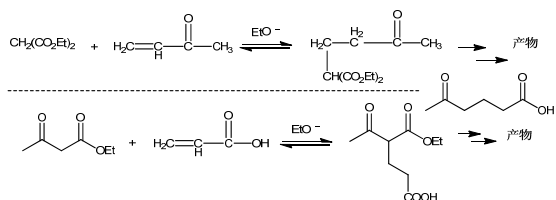
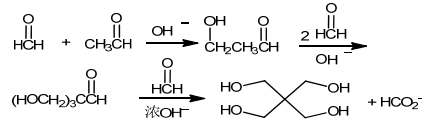
1) 分别用三乙和丙二酸为原料, 合成下列化合物



2) 以乙醛为原料, 利用羟醛缩合和歧化反应合成下列化合物



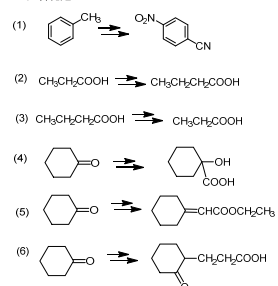
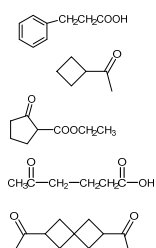
24

1) 分别用三乙和丙二酸为原料, 合成下列化合物 **测试答案**2) 以乙醛为原料, 利用**羟醛缩合**和**歧化反应**合成下列化合物

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作业

1. 转化题

2. 以苯、甲苯、C2及以下有机物合成下列物质
(三乙或丙二酸酯方法)

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